

FOR TEACHERS ONLY

The University of the State of New York
REGENTS HIGH SCHOOL EXAMINATION

PHYSICAL SETTING/CHEMISTRY

Wednesday, August 20, 2025 — 8:30 to 11:30 a.m., only

RATING GUIDE

Directions to the Teacher:

Refer to the directions on page 2 before rating student papers.

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: <https://www.nysed.gov/state-assessment/high-school-regents-examinations> and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

Directions to the Teacher

Follow the procedures below for scoring student answer papers for the Regents Examination in Physical Setting/Chemistry. Additional information about scoring is provided in the publication *Information Booklet for Scoring Regents Examinations in the Sciences*.

At least two science teachers must participate in the scoring of the Part B–2 and Part C open-ended questions on a student's paper. Each of these teachers should be responsible for scoring a selected number of the open-ended questions on each answer paper. No one teacher is to score more than approximately one-half of the open-ended questions on a student's answer paper. Teachers may not score their own students' answer papers.

Students' responses must be scored strictly according to the Rating Guide. For open-ended questions, credit may be allowed for responses other than those given in the rating guide if the response is a scientifically accurate answer to the question and demonstrates adequate knowledge, as indicated by the examples in the rating guide. Do not attempt to correct the student's work by making insertions or changes of any kind. On the student's separate answer sheet, for each question, record the number of credits earned and the teacher's assigned rater/scorer letter.

Fractional credit is *not* allowed. Only whole-number credit may be given for a response. If the student gives more than one answer to a question, only the first answer should be rated. Units need not be given when the wording of the questions allows such omissions.

For hand scoring, raters should enter the scores earned in the appropriate boxes printed on the separate answer sheet. Next, the rater should add these scores and enter the total in the box labeled "Total Raw Score." Then the student's raw score should be converted to a scale score by using the conversion chart that will be posted on the Department's web site at: <https://www.nysed.gov/state-assessment/high-school-regents-examinations> on Wednesday, August 20, 2025. The student's scale score should be entered in the box labeled "Scale Score" on the student's answer sheet. The scale score is the student's final examination score.

Schools are not permitted to rescore any of the open-ended questions on this exam after each question has been rated once, regardless of the final exam score. Schools are required to ensure that the raw scores have been added correctly and that the resulting scale score has been determined accurately.

Because scale scores corresponding to raw scores in the conversion chart may change from one administration to another, it is crucial that, for each administration, the conversion chart provided for that administration be used to determine the student's final score.

Part B–2

Allow a total of 15 credits for this part. The student must answer all questions in this part.

51 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The average kinetic energy decreases.

The average KE goes down.

decreases

52 [1] Allow 1 credit for any value from 80.°C to 82°C, inclusive.

53 [1] Allow 1 credit for 9470 J *or* for any value from 9470 J to 9500 J, inclusive.

54 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

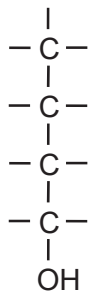
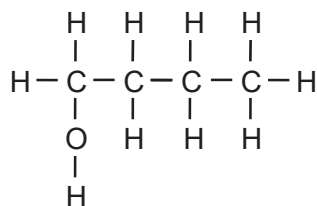
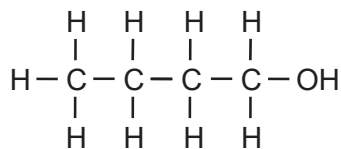
Ethane molecules have only single carbon-carbon bonds.

There are no multiple bonds between the carbon atoms.

No carbon-to-carbon double or triple bonds.

55 [1] Allow 1 credit.

Examples of 1-credit responses:



Note: Do *not* allow credit if only some of the H atoms bonded to C atoms are shown.
Organic functional groups must be completely shown.

56 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The boiling point of each successive alcohol increases as the number of carbon atoms increases.

Boiling points increase.

more carbons, higher boiling point

increases

57 [1] Allow 1 credit for 0 *or* zero.

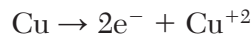
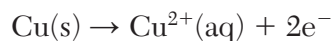
58 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

chemical potential energy

chemical

potential

59 [1] Allow 1 credit. Acceptable responses include, but are not limited to:



Note: Do *not* allow credit for the e without the minus sign (−).

60 [1] Allow 1 credit for blue.

61 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The HCl(aq) solution contains mobile ions.

The H^{+} and Cl^{-} ions in the solution are moving.

The solution has aqueous ions.

62 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

$$M_A (10.0 \text{ mL}) = (0.10\text{M})(20.0 \text{ mL})$$

$$\frac{(0.10\text{M})(20.0 \text{ mL})}{10.0 \text{ mL}}$$

$$\frac{(.10)(20)}{10}$$

63 [1] Allow 1 credit for 12.0 *or* 12.

64 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

In this reaction, an isotope of phosphorus is changing into an isotope of silicon.

A different element is formed.

Phosphorous has an atomic number of 15 and becomes silicon with an atomic number of 14.

65 [1] Allow 1 credit for 10.00 min. Significant figures do *not* need to be shown.

Part C

Allow a total of 20 credits for this part. The student must answer all questions in this part.

- 66** [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The energy of an electron in the fourth shell is higher than the energy of an electron in the first shell.

The fourth shell electron has greater energy.

The electron in the first shell has less.

- 67** [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The spectral lines seen in the spectroscope are compared to the known bright-line spectrum of gallium.

The spectrum from the sample is matched to gallium on a chart of element spectra.

Compare the spectral wavelengths to those of gallium.

- 68** [1] Allow 1 credit. Acceptable responses include, but are not limited to:

$$(68.926 \text{ u})(0.6011) + (70.925 \text{ u})(0.3989)$$

$$\frac{60.11(68.926) + 70.925 (39.89)}{100}$$

$$39.89\%(70.925) + 60.11\%(68.926)$$

Note: Do *not* allow credit for a numerical setup using mass numbers rather than isotopic masses.

- 69** [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Both atoms have 31 protons. Ga-69 has 38 neutrons and Ga-71 has 40 neutrons.

same number of protons, different number of neutrons

- 70** [1] Allow 1 credit. Acceptable responses include, but are not limited to:

Two reactants form one product.

One substance is formed from two substances.

Two compounds form one compound.

- 71 [1] Allow 1 credit for 82.4% *or* for any value from 82% to 82.412%, inclusive.
- 72 [1] Allow 1 credit for 98 g/mol *or* for any value from 97.996 g/mol to 98.12 g/mol, inclusive.
- 73 [1] Allow 1 credit for 2 NH₃ + _____ H₂SO₄ → _____ (NH₄)₂SO₄
 Allow credit even if the coefficient “1” is written in front of H₂SO₄ and/or (NH₄)₂SO₄.
- 74 [1] Allow 1 credit. Acceptable responses include, but are not limited to:
 Sodium is a metal and chlorine is a nonmetal, so the bonding is ionic.
 An active metal reacted with an active nonmetal.
- 75 [1] Allow 1 credit for Ar *or* argon.
- 76 [1] Allow 1 credit. The positions of the electrons may vary.

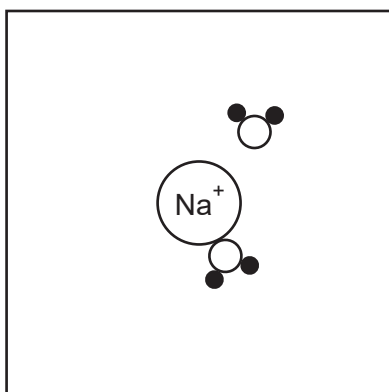
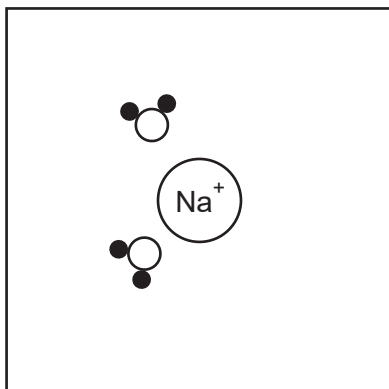
Examples of 1-credit responses:



- 77 [1] Allow 1 credit. Acceptable responses include, but are not limited to:
 The Mg²⁺ ion is smaller because it has two shells of electrons and a Mg atom has three shells of electrons.
 The Mg²⁺ ion forms when the Mg atom loses two electrons.
 The magnesium ion has 10 electrons and the magnesium atom has 12 electrons.
 A Mg atom is formed when the Mg²⁺ gains two electrons.
- Note:** Do *not* allow credit for a response indicating that the Mg²⁺ ion lost electrons.

- 78 [1] Acceptable responses must show at least two water molecules. The oxygen atom of each water molecule must face toward the Na^+ ion.

Examples of 1-credit responses:



- 79 [1] Allow 1 credit for unsaturated *or* not saturated.

- 80 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

From $30.^{\circ}\text{C}$ to $70.^{\circ}\text{C}$, the solubility of KCl in water increases.

At 70°C , KCl is more soluble than at 30°C .

The solubility increases.

increases

- 81 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

At standard pressure, the boiling point of the solution is higher than the boiling point of water.

The boiling point of water is lower.

82 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

The rates of the forward and reverse reactions are equal.

The two reactions occur at the same rate.

equal

same

83 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

177 800 J

1.778×10^5 J

84 [1] Allow 1 credit. Acceptable responses include, but are not limited to:

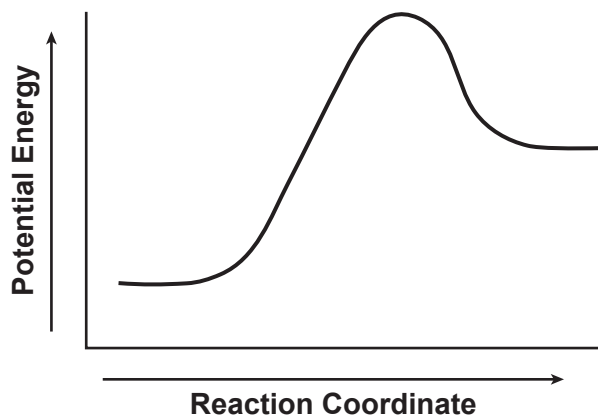
A greater number of effective collisions occur between CO_2 and CaO , producing more CaCO_3 .

More CO_2 molecules collide with CaO , producing more CaCO_3 .

Adding CO_2 increases the number of collisions between CaO and CO_2 .

85 [1] Allow 1 credit for showing that the PE of the products is higher than the PE of the reactants.

Example of a 1-credit response:



The *Chart for Determining the Final Examination Score for the August 2025 Regents Examination in Physical Setting/Chemistry* will be posted on the Department's web site at: <https://www.nysed.gov/state-assessment/high-school-regents-examinations> on Wednesday, August 20, 2025. Conversion charts provided for previous administrations of the Regents Examination in Physical Setting/Chemistry must NOT be used to determine students' final scores for this administration.

Online Submission of Teacher Evaluations of the Test to the Department

Suggestions and feedback from teachers provide an important contribution to the test development process. The Department provides an online evaluation form for State assessments. It contains spaces for teachers to respond to several specific questions and to make suggestions. Instructions for completing the evaluation form are as follows:

1. Go to <https://www.nysed.gov/state-assessment/teacher-feedback-state-assessments>.
2. Click Regents Examinations.
3. Complete the required demographic fields.
4. Select the test title from the Regents Examination dropdown list.
5. Complete each evaluation question and provide comments in the space provided.
6. Click the SUBMIT button at the bottom of the page to submit the completed form.

Map to Core Curriculum

August 2025 Physical Setting/Chemistry			
Question Numbers			
Key Ideas/Performance Indicators	Part A	Part B	Part C
Standard 1			
Math Key Idea 1		38	68, 69, 83
Math Key Idea 2		51, 52, 56, 59, 62	73
Math Key Idea 3		32, 36, 39, 53, 65	71, 72
Science Inquiry Key Idea 1		40, 41, 43, 44, 48, 49, 50, 51, 54, 57, 58, 60, 61, 63, 64	66, 67, 70, 74, 75, 76, 77, 79, 80, 81, 82, 84
Science Inquiry Key Idea 2			
Science Inquiry Key Idea 3		34, 35, 38, 40, 41, 42, 43, 44, 46, 48, 51, 52, 55, 59, 60, 64	70, 75, 78, 80
Engineering Design Key Idea 1			
Standard 2			
Key Idea 1			
Key Idea 2			
Key Idea 3			
Standard 6			
Key Idea 1			
Key Idea 2		55, 76, 78, 85	
Key Idea 3		47, 63	
Key Idea 4			
Key Idea 5		56, 80	
Standard 7			
Key Idea 1			
Key Idea 2			
Standard 4 Process Skills			
Key Idea 3		31, 32, 33, 36, 37, 40, 42, 44, 45, 47, 54, 59, 60, 62, 63	67, 68, 69, 72, 73, 79, 80, 82, 84
Key Idea 4		51, 52, 53, 65	85
Key idea 5		38	75, 76
Standard 4			
Key Idea 3	1, 2, 3, 4, 5, 7, 8, 12, 13, 14, 15, 16, 17, 18, 19, 20, 22, 23, 24, 25, 26, 27, 28, 29, 30	31, 32, 33, 34, 36, 37, 40, 41, 42, 43, 44, 45, 46, 47, 48, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63	66, 67, 68, 69, 70, 71, 72, 73, 79, 80, 81, 82, 84
Key Idea 4	21	49, 51, 52, 53, 65	83, 85
Key Idea 5	6, 9, 10, 11	35, 38, 39, 50, 64	74, 75, 76, 77, 78
Reference Tables			
2011 Edition	4, 5, 9, 11, 13, 24, 26, 27, 28, 29, 30	31, 33, 34, 35, 37, 38, 39, 40, 43, 49, 53, 55, 62	66, 71, 72, 74, 75, 76, 77, 79